Problem Set 5B

Intersection of lines and skew lines

Do these after the slides for Section 5B. Textbook Exercise 5B. Answers are on the last page. Show full working in your exercise book.

Skills

1 Show that the lines $\mathbf{r} = \mathbf{i} + \mathbf{j} + \lambda(\mathbf{i} + 2\mathbf{j} + \mathbf{k})$ and $\mathbf{r} = -2\mathbf{i} + 5\mathbf{j} - 5\mathbf{k} + \mu(2\mathbf{i} - \mathbf{j} + 3\mathbf{k})$ intersect, and find the point of intersection.

2 Show that the lines $\mathbf{r} = \mathbf{j} + 2\mathbf{k} + \lambda(\mathbf{i} + \mathbf{j})$ and $\mathbf{r} = \mathbf{i} + \mathbf{k} + \mu(\mathbf{j} + \mathbf{k})$ are skew.

3 Describe the relationship between the lines $\mathbf{r} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k} + \lambda(2\mathbf{i} - \mathbf{j} + 4\mathbf{k})$ and $\mathbf{r} = \mathbf{j} + \mathbf{k} + \mu(-4\mathbf{i} + 2\mathbf{j} - 8\mathbf{k})$.

4 Find, correct to one decimal place, the acute angle between the two lines in Question 1.

5 Find, correct to one decimal place, the acute angle between the lines $\dfrac{x-1}{2} = \dfrac{y}{2} = z + 1$ and $\mathbf{r} = \mathbf{j} + \mu(\mathbf{i} - \mathbf{k})$.

Exam style

6 The lines $\mathbf{r} = \mathbf{i} + 2\mathbf{k} + \lambda(\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ and $\mathbf{r} = 2\mathbf{i} + k\mathbf{j} - 2\mathbf{k} + \mu(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$ intersect. Find k and the point of intersection.

7 Two particles move so that at time t their positions are $\mathbf{r}_A = \mathbf{i} + 2\mathbf{j} + t(\mathbf{i} + \mathbf{j} + 2\mathbf{k})$ and $\mathbf{r}_B = 2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k} + t(\mathbf{i} - 2\mathbf{j} + \mathbf{k})$. Show that their paths cross but the particles do not collide.

8 The lines $\mathbf{r} = \mathbf{i} + \mathbf{j} + \mathbf{k} + \lambda(a\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ and $\mathbf{r} = 2\mathbf{i} + 3\mathbf{k} + \mu(3\mathbf{i} + a\mathbf{j} + 5\mathbf{k})$ are perpendicular. Find a, and determine whether the lines intersect.

Problem Set 5B: answers

1 $\lambda = 1,\ \mu = 2$: $(2, 3, 1)$

2 x and z give $\lambda = 1,\ \mu = 1$, but then y: $2 \neq 1$; directions not parallel

3 Parallel and distinct

4 $\cos\theta = \dfrac{3}{\sqrt{84}}$, $\theta = 70.9°$

5 $\cos\theta = \dfrac{1}{3\sqrt{2}}$, $\theta = 76.4°$

6 $\mu = 1,\ \lambda = 2$: $k = 5$, point $(3, 4, 0)$

7 Paths meet at $(3, 4, 4)$: A is there at $t = 2$, B at $t = 1$

8 $3a + 2a - 5 = 0$, $a = 1$; no common point, so they are skew